

Executive Productivity Agent — Assignment 1

Turning messy executive inputs into a daily action brief

AGENTIC AI FACTORY — ASSIGNMENT 1

USER: ARJUN MALHOTRA, VP SALES

Concise demo: product design, architecture, pipeline, and an example daily brief for an enterprise executive.

Made by Arnav Tayal | Enrollment: E23CSEU1556

Problem statement

Executives receive commitments across meetings, email, calendars and messages — information is scattered and inconsistent.

Dates are relative or vague ("tomorrow", "next week") and often conflict with calendar context.

Missed or unclear actions. Executives need a concise daily brief answering: What did I promise? What needs action today? Who owns what?

Key takeaway: Reduce fragmentation by converting noisy inputs into a single, actionable daily brief.

What the agent does

Identify commitments

Extract promises and action statements from transcripts, emails and chats.

Classify ownership

Separate "My Actions" vs "Waiting on Others" and flag unclear ownership.

Detect deadlines

Resolve absolute and relative dates against calendar context.

Deduplicate

Merge duplicated actions appearing across multiple inputs.

Flag ambiguity

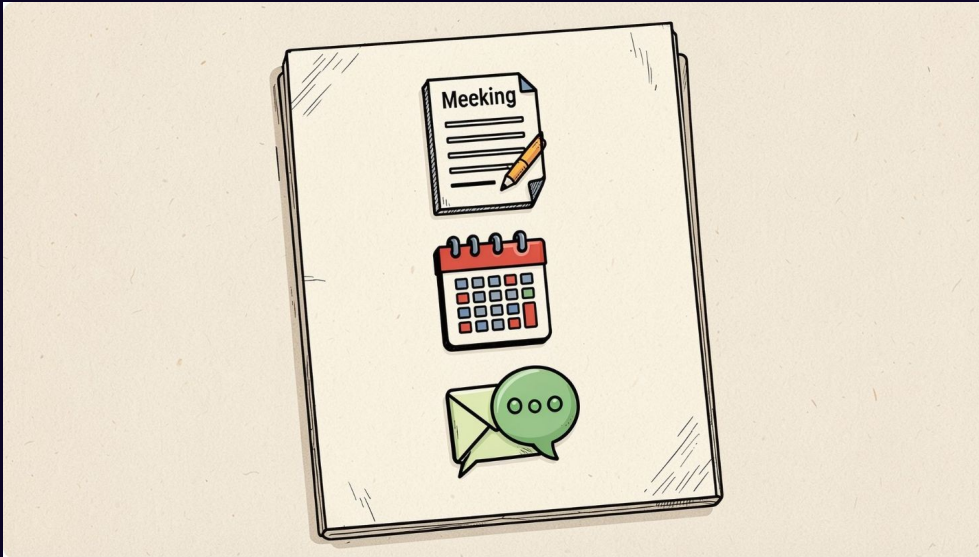
Highlight unclear ownership or vague commitments instead of guessing.

Generate brief & Q&A

Produce a concise Executive Daily Brief and support natural-language queries.

Key takeaway: End-to-end conversion from noisy inputs to an actionable brief and interactive Q&A.

Inputs & source data



Source categories

- Meeting transcripts
- Executive calendar (Arjun)
- Other people's calendars
- Emails, messages and shared documents

Important people (data sources)

- Arjun Malhotra — VP Sales (agent user)
- Neha Kapoor — Marketing Lead
- Raghav Sethi — Ops Manager
- Divya Rao — Finance
- Priya Nair — Meridian Logistics (external)
- Facilities — internal distribution list

Note: The agent reads these sources on Arjun's behalf; Arjun is the primary user and decision-maker.

Key takeaway: The agent merges structured (calendars) and unstructured (transcripts, email) sources to form a unified action model.

Agent architecture



Architecture combines parsing (NLP), symbolic normalization (date resolution, calendar checks) and deterministic validation steps before brief generation.

Key takeaway: A hybrid pipeline (ML + rule-based) ensures traceable transformations and avoids hallucinated ownership.

Core reasoning & process flow

01

1. Commitments extraction

Detect verbs, promises and obligations in text.

03

3. Deadline resolution

Normalize relative dates (e.g., "tomorrow") to calendar dates.

05

5. Deduplication & classification

Merge duplicates and classify: My Action / Waiting on Others / Unclear.

02

2. Attribution

Identify who made the commitment and intended recipient.

04

4. Calendar validation

Cross-check commitments with Arjun's calendar and attendees' calendars.

06

6. Prioritization & brief generation

Rank tasks for today's brief and emit clarifying prompts for ambiguous items.



Agent flags ambiguity rather than inventing ownership; clarifying prompts surface to Arjun for confirmation.

Key takeaway: Deterministic checks and explicit clarification rules prevent hallucination and preserve executive control.

Example: from messy input to action

Raw transcript

"Arjun: I'll send Raghav the updated vendor list by end of day Wednesday."

Agent extraction

- Commitment: Send updated vendor list
- Owner: Arjun Malhotra
- Recipient: Raghav Sethi
- Deadline (normalized): Wednesday, 23 September 2026
- Category: My Action — Status: Pending

Ambiguous example

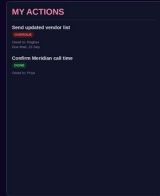
"Mumbai office renewal paperwork needs someone to sign off this week."

Agent behaviour

- Extract action: Mumbai office renewal — sign off
- Ownership: **Unclear** — agent flags for clarification
- Output: Needs clarification card (do not assign Facilities or another person)

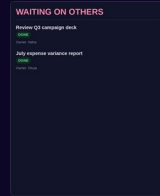
Key takeaway: The agent reliably transforms explicit commitments and surface ambiguous items for human clarification.

Daily executive brief (example)



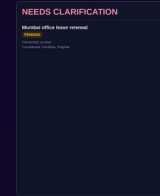
MY ACTIONS

- Send updated vendor list to Raghav — due Wednesday
- Reconfirm Meridian Logistics call time
- Review Q3 campaign deck
- Prepare board prep items (as applicable)



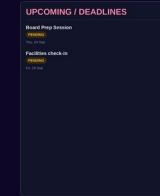
WAITING ON OTHERS

- Q3 campaign deck — Neha
- July expense variance report — Divya



NEEDS CLARIFICATION

- Mumbai office renewal paperwork — ownership unclear



UPCOMING / DEADLINES

- Campaign deck review — Thursday morning (safer commitment)
- Board Prep Session — Thursday, 24 September
- Facilities check-in — Friday, 25 September

Key takeaway: Concise, card-based brief showing only actions derived from provided inputs; ambiguous ownership is not assumed.

Q&A / agent interaction

ASK THE AGENT

What did I promise Raghav?

✓ You promised Raghav the following:

You committed to sending the updated vendor list by end of day Wednesday, 23 September.

What needs action today?

✓ Today's actions:

Vendor list to Raghav — overdue since Wed, 23 Sep. Meridian call time — pending.

Example queries

- **Q:** "What did I promise Raghav?"
A: "You committed to sending Raghav the updated vendor list by end of day Wednesday, 23 September."
- **Q:** "What needs action today?"
A: Agent returns today's My Actions filtered by deadline and calendar conflicts.

The agent answers using extracted commitments, normalized deadlines and calendar checks — not by summarizing unrelated content.

Key takeaway: Interactive, explainable responses sourced from traceable extractions and calendar validation.

Value, AI tools & next steps

Deliverables: Architecture + process flow, working agent / clickable prototype, and demo assets.

- [GitHub Repository]
- [Demo Video]



GitHub Repository

Source code, README & setup instructions

[LINK TO BE ADDED](#)



Demo Video

Recorded walkthrough of the live agent

[LINK TO BE ADDED](#)

Next steps for demo

1. Live walkthrough of ingestion → brief generation
2. Interactive Q&A with the agent
3. Clarification flow: resolve 'Needs clarification' items

Business value

- Reduces information overload
- Makes commitments explicit and actionable
- Separates personal work from dependencies
- Detects deadlines and prevents duplicates
- Surfaces ambiguous ownership for clarification

AI tools used

- LLM for extraction & Q&A
- Rule-based date normalization & calendar validation
- Deterministic deduplication and classification logic
- AI-assisted development for prototyping and UI generation

Key takeaway: The agent converts noisy executive data into transparent, actionable decisions while preserving human oversight and preventing hallucination.